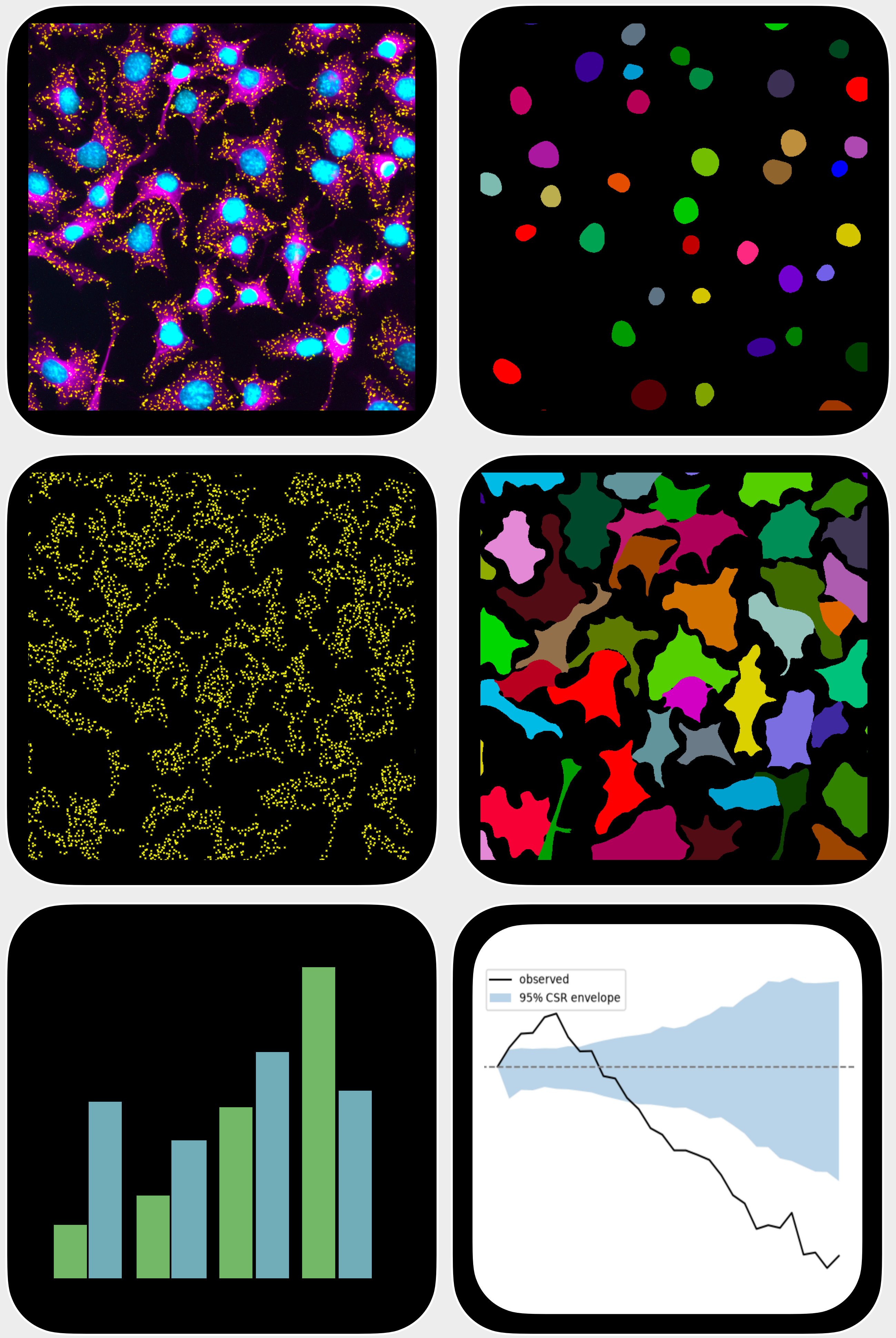
Welcome to the

BoBiAC

Boston Bioimage Analysis Course | 2026

CITE: cite.hms.harvard.edu



Date: 6th -11th July 2025, 9:00 am - 6:30 pm

Time: 9:00 am - 6:30 pm

location: Gordon Hall - Harvard Medical School

website: bobiac.github.io

course material: bobiac.github.io/bobiac-book

jupyter

Image Analysis Collaboratory

CITE
Core for Imaging Technology & Education

BioImaging North America

HARVARD MEDICAL SCHOOL

The BoBiAC Team

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Amelie Andreas
Research Assistant
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Harvard Medical School

Organizer & Instructor Teaching Assistant

ℹ️ General Information

📞 **Internal communications**:
- Slack channels (mainly #course-announcements & DMs)

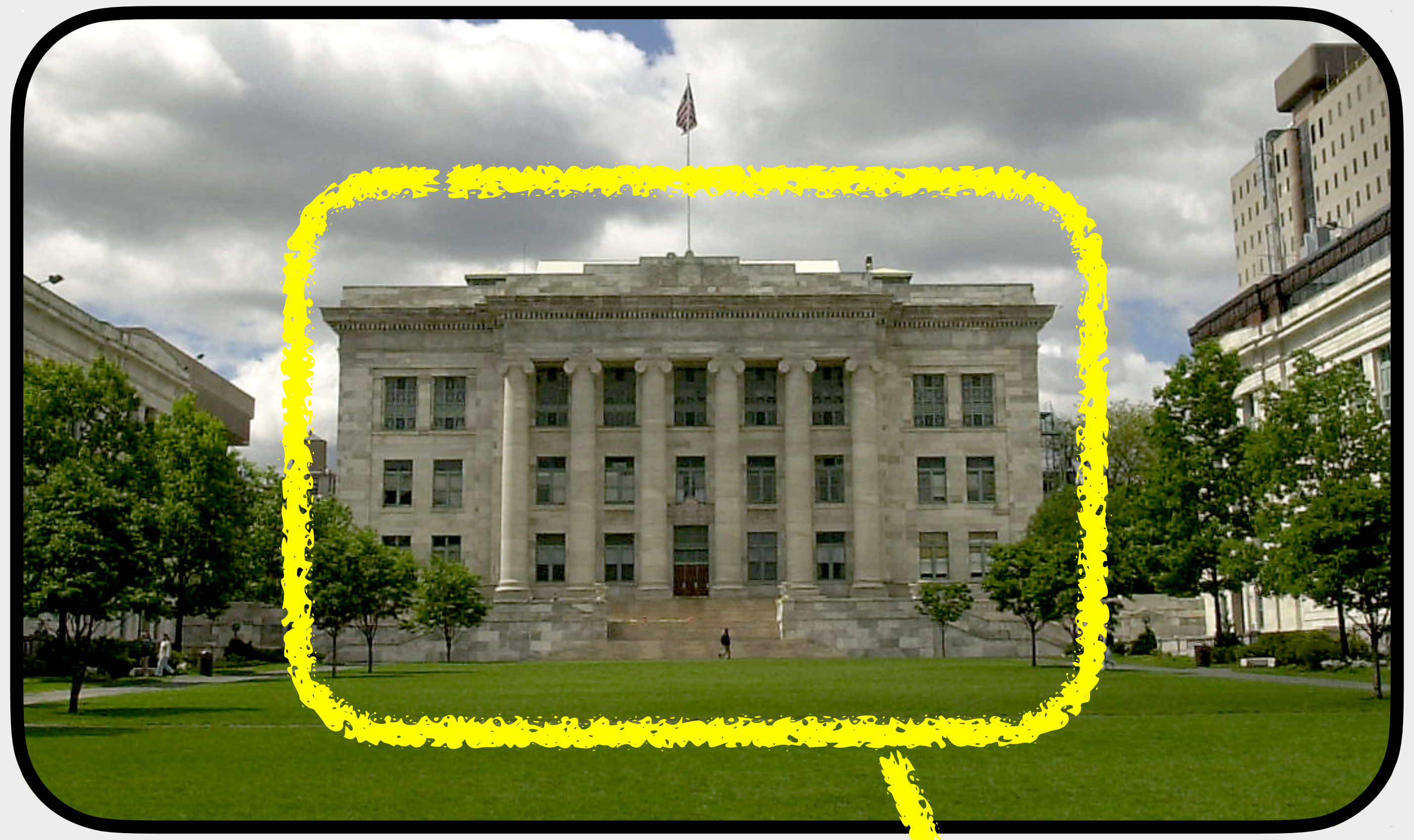
📍**location**:
- Harvard Medical School - Gordon Hall - Room 106

⌚ **time**:
- 9:00 am - 6.30 pm
- optional office hours from 7:30 pm

☕ **breakfast**:
- bagels 🥯, coffee, tea & snacks @ 9:00 am

🍕 **lunch**:
- 12:00 pm in Gordon Hall, Room 106

🍔 **social dinner**:
- Thursday July 9th @ 7:00 pm
- Shy Bird (201 Brookline Ave, Boston - 10 min walk)
- we leave @ 6:30 pm (🚶)



🌐 WiFi Connection

- **Harvard Secure**: if you have Harvard credentials
- **Eduroam**: if you have academic credential (edu)
- **Harvard Guest**: you will need to register your laptop for that (if you provided the MAC address in advance, you should be able to directly connect)
- If all of above do not work, let us know!

Course Structure

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- The course is designed for **beginners** in python and image analysis
- Each day consists of a **mix of lectures** explaining key bioimage analysis concepts, interspersed with **practical, hands-on exercises** using **Python**.
- Every morning, we will **start** the day **with coffee** and **open discussion** (Q&A).
- Most of these **exercises** will be completed either **step-by-step as a class** or in small groups.
- The course should be **interactive**, there are absolutely **no stupid questions**, you are encouraged to ask questions. During the practical exercise feel free to consult with your neighbors, you can help each other to understand concept.
- Each day (first four days), five participants will give a **brief 3–5 minute introduction** using their prepared slide (introduce yourself, your work, and the image analysis tasks you hope to perform).

Schedule

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<table>
<tr><th></th><th>Monday 6th July</th><th>Tuesday 7th July</th><th>Wednesday 8th July</th><th>Thursday 9th July</th><th>Friday 10th July</th><th>Saturday 11th July</th></tr>
<tr><td>9:00am - 9:30am</td><td>Welcome + Intro</td><td>Coffee & Questions?</td><td>Coffee & Questions?</td><td>Coffee & Questions?</td><td>Coffee & Questions?</td><td>Coffee & Questions?</td></tr>
<tr><td>9:30am - 10:00am</td><td>Coffee Break</td><td>Error Messages</td><td rowspan="3">Segmentation (Machine Learning)</td><td rowspan="3">Segmentation (Deep Learning)</td><td rowspan="3">Single Cell Measurements</td><td rowspan="3">Cell Population Measurements (Spatial Statistics & Colocalization)</td></tr>
<tr><td>10:00am - 10:30am</td><td rowspan="2">Getting Started with Python</td><td rowspan="2">Python Bioimage Analysis</td></tr>
<tr><td>10:30am - 11:00am</td></tr>
<tr><td>11:00am - 11:15am</td><td rowspan="2">The Python Basics</td><td>Coffee Break</td><td>Coffee Break</td><td>Coffee Break</td><td>Coffee Break</td><td>Coffee Break</td></tr>
<tr><td>11:15am - 11:30am</td><td rowspan="2">Segmentation (Classic)</td><td rowspan="2">Segmentation (Deep Learning)</td><td rowspan="2">Deep Learning Spot Detection</td><td rowspan="2">Single Cell Measurements</td><td rowspan="2">Cell Population Measurements (Spatial Statistics & Colocalization)</td></tr>
<tr><td>11:30am - 12:00pm</td><td>Laptops Setup</td></tr>
<tr><td>12:00pm - 12:30pm</td><td rowspan="2">Lunch</td><td rowspan="2">Lunch</td><td rowspan="2">Lunch</td><td rowspan="2">Lunch</td><td rowspan="2">Lunch</td><td rowspan="2">Lunch</td></tr>
<tr><td>12:30pm - 1:00pm</td></tr>
<tr><td>1:00pm - 1:30pm</td><td>Student Presentations (5x5min)</td><td>Student Presentations (5x5min)</td><td>Student Presentations (5x5min)</td><td>Student Presentations (5x5min)</td><td rowspan="5">Cell Population Measurements (Spatial Statistics & Colocalization)</td><td rowspan="3">Student Group Work</td></tr>
<tr><td>1:30pm - 2:00pm</td><td rowspan="2">The Python Basics</td><td rowspan="4">Segmentation (Classic)</td><td rowspan="4">Segmentation (Deep Learning)</td><td rowspan="2">Managing your own Python Workflow</td></tr>
<tr><td>2:00pm - 2:30pm</td></tr>
<tr><td>2:30pm - 3:00pm</td><td rowspan="2">Introduction to Digital Images</td><td rowspan="2">Student Group Work</td><td rowspan="2">How to Acquire Analyzable Microscopy Images</td></tr>
<tr><td>3:00pm - 3:30pm</td></tr>
<tr><td>3:30pm - 4:00pm</td><td>Coffee Break</td><td>Coffee Break</td><td>Coffee Break</td><td>Coffee Break</td><td>Coffee Break</td><td>Coffee Break</td></tr>
<tr><td>4:00pm - 4:30pm</td><td rowspan="3">Python for Bioimage Analysis</td><td rowspan="3">Segmentation (Machine Learning)</td><td rowspan="3">Segmentation (Deep Learning)</td><td rowspan="2">Student Group Work</td><td rowspan="3">Cell Population Measurements (Spatial Statistics & Colocalization)</td><td rowspan="3">Using Python in the Real World</td></tr>
<tr><td>4:30pm - 5:00pm</td></tr>
<tr><td>5:00pm - 5:30pm</td><td>Measurements and Quantification</td></tr>
<tr><td>5:30pm - 6:00pm</td><td>Coffee Break</td><td>Coffee Break</td><td>Coffee Break</td><td>Coffee Break</td><td>Coffee Break</td><td>Coffee Break</td></tr>
<tr><td>6:00pm - 6:30pm</td><td>Python for Bioimage Analysis</td><td>Segmentation (Machine Learning)</td><td>Segmentation (Deep Learning)</td><td>Measurements and Quantification</td><td>Measurements and Quantification</td><td>Feedback & Wrap-Up</td></tr>
<tr><td>6:30pm - 7:30pm</td><td></td><td></td><td></td><td rowspan="3">Social Dinner</td><td></td><td rowspan="3"></td></tr>
<tr><td>7:30pm - 8:30pm</td><td>Optional: Office Hour</td><td>Optional: Office Hour</td><td>Optional: Office Hour</td><td>Optional: Office Hour</td></tr>
<tr><td>8:30pm - 9:30pm</td><td></td><td></td><td></td><td></td></tr>
</table>

📋 Course Topics I

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Learning Python Basics

- **Getting Started with Python and uv**: what is python? How do I install it? How do I use it?
- **The Python Basics**: how do I write python code? What is the syntax?

📋 Course Topics II

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Learning Python for Bioimage Processing & Analysis

- **Digital Images & Python**: what is a digital image? How do I deal with it in python?
- **Image Segmentation with Python**: what are semantic and instance segmentation? how do I perform segmentation on my fluorescence images? [Classical Methods, Ilastik (ML) & Cellpose (DL)]
- **Spot Detection**: how can we detect small spot-like objects such as fluorescent puncta, foci, beads, or RNA spots in fluorescence images? [Spotiflow (DL)]
- **Object Classification**: how can I classify objects in my images into different categories (e.g., mitotic vs. non-mitotic cells) to enable class-specific analysis? [Ilastik]
- **Measurements & Quantification with Python**: how can I extract quantitative data from my fluorescence images for plotting and drawing conclusions from my experiments?
- **Colocalization analysis with Python**: what is colocalization in fluorescence microscopy? How can I quantify it?

📋 Course Topics II

instance segmentation

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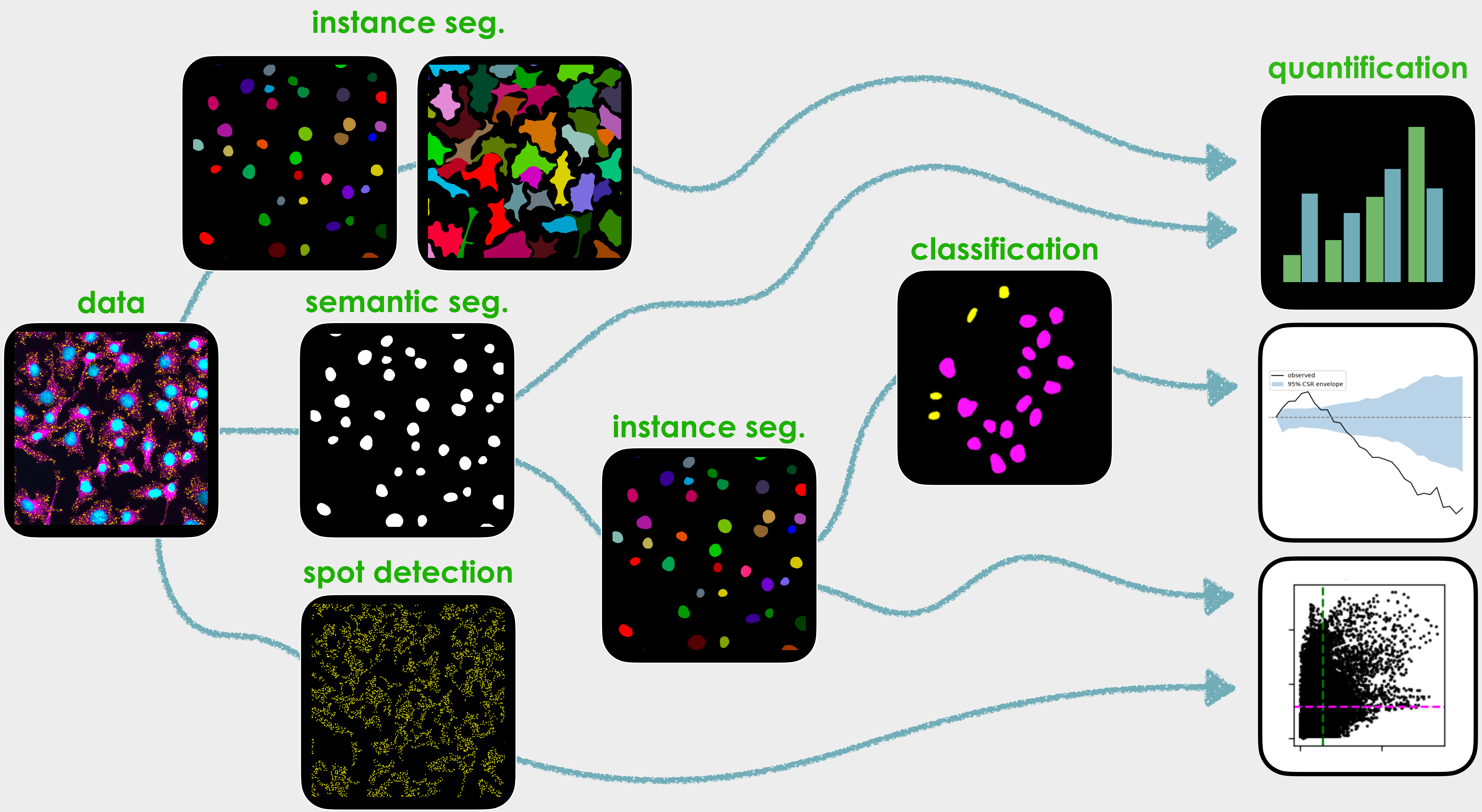
Learning Python for Bioimage Processing & Analysis



🎓 Learning Objectives

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- ~~Become a **Python & Bioimage Analysis Expert!**~~

🎓 Learning Objectives

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- ~~Become a **Python & Bioimage Analysis Expert!**~~
- Gain a basic understanding of **what Python is**.
- Learn how to **get started with Python**: **installing Python**, setting up **virtual environments**, and launching **Jupyter Notebooks** and **.py** files.
- Get familiar with key **Python packages** commonly used in bioimage analysis.
- Learn how to **load**, **handle**, and **display images** using Python.
- Explore different approaches to **image segmentation** in Python, including classical methods, machine learning (Ilastik), and deep learning (Cellpose).
- Learn how to use deep learning for spot detection (Spotiflow).
- Understand the basics of **image quantification and measurements** and how to apply them using Python (e.g. single-cell measurements, spatial statistics, and colocalization).

(‣Improve LLM prompts)

📒 Course Material

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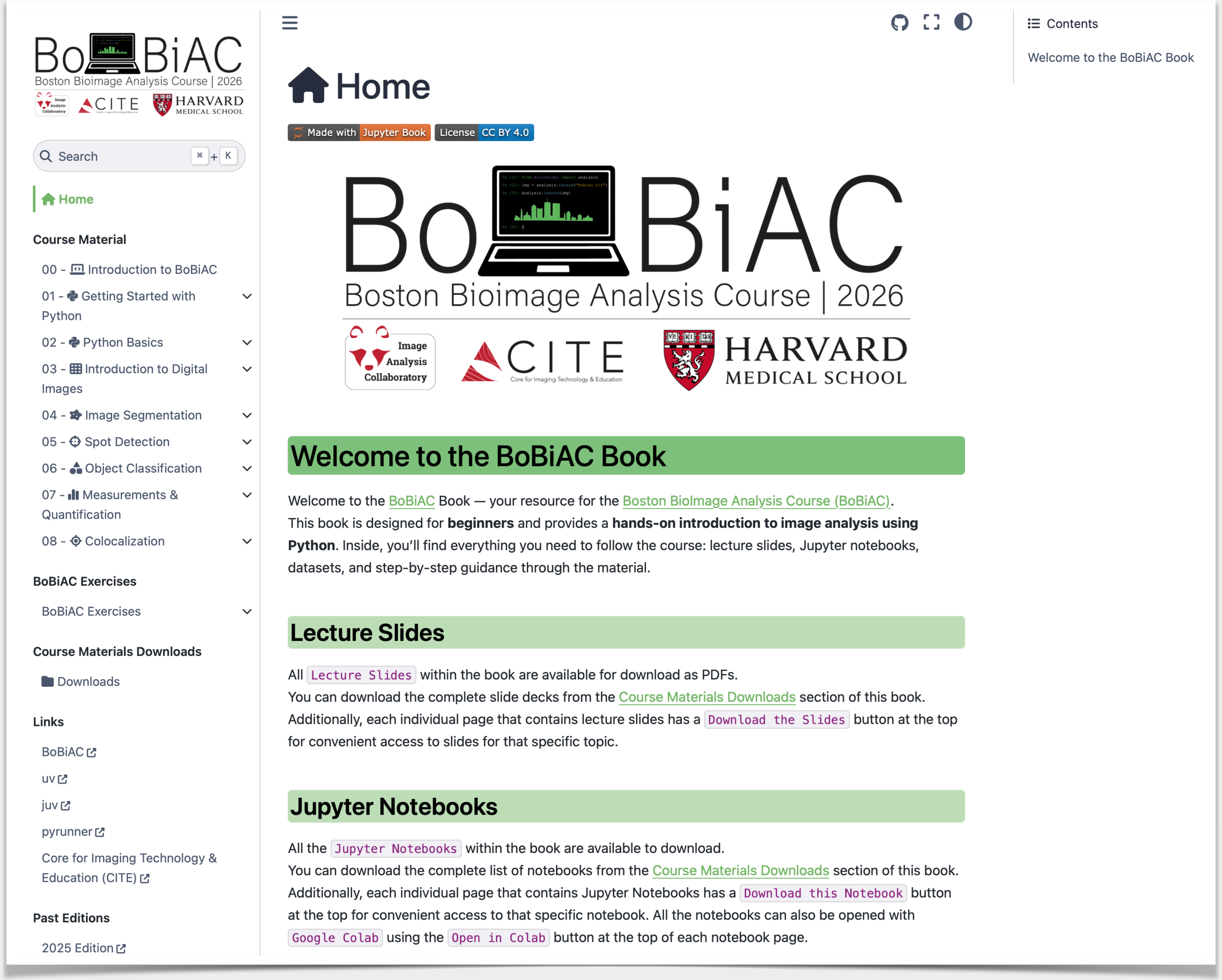

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Search ⌘ + K

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- 04 - Image Segmentation
- 05 - Spot Detection
- 06 - Object Classification
- 07 - Measurements & Quantification
- 08 - Colocalization

BoBiAC Exercises

- BoBiAC Exercises

Course Materials Downloads

- Downloads

Links

- BoBiAC
- uv
- juv
- pyrunner
- Core for Imaging Technology & Education (CITE)

Past Editions

- 2025 Edition

Home

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Welcome to the BoBiAC Book

Welcome to the BoBiAC Book — your resource for the Boston BioImage Analysis Course (BoBiAC). This book is designed for **beginners** and provides a **hands-on introduction to image analysis using Python**. Inside, you'll find everything you need to follow the course: lecture slides, Jupyter notebooks, datasets, and step-by-step guidance through the material.

Lecture Slides

All `Lecture Slides` within the book are available for download as PDFs.
You can download the complete slide decks from the Course Materials Downloads section of this book.
Additionally, each individual page that contains lecture slides has a `Download the Slides` button at the top for convenient access to slides for that specific topic.

Jupyter Notebooks

All the `Jupyter Notebooks` within the book are available to download.
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Contents

Welcome to the BoBiAC Book

🔗 bobiac.github.io/bobiac-book

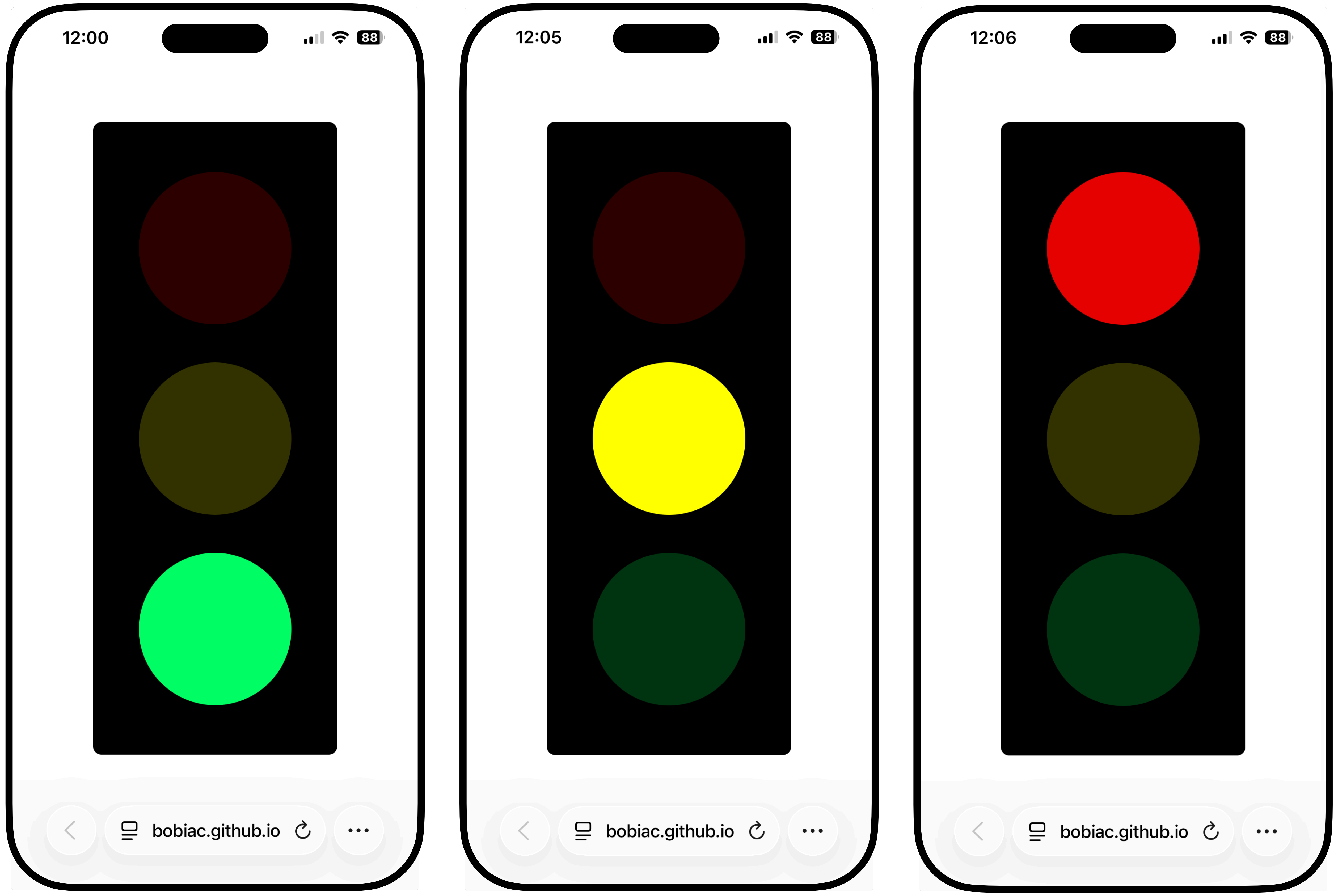
🚦 classlight

By Talley Lambert

🔗 bobiac.github.io

bobiac.github.io/classlight/bobiac2026

Realtime Feedback!



? Questions

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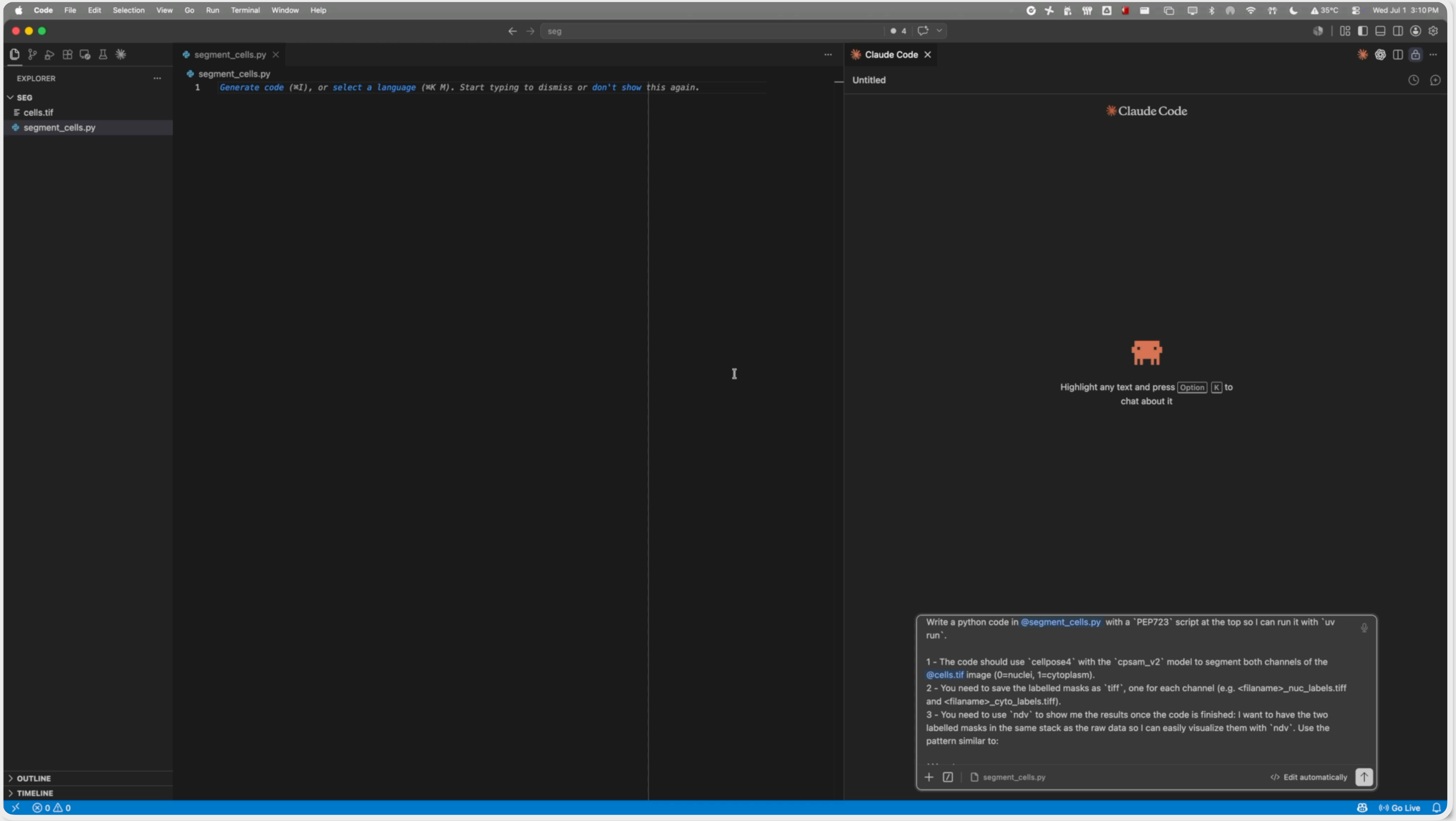
Why learn image analysis/Python if we have AI/LLMs?



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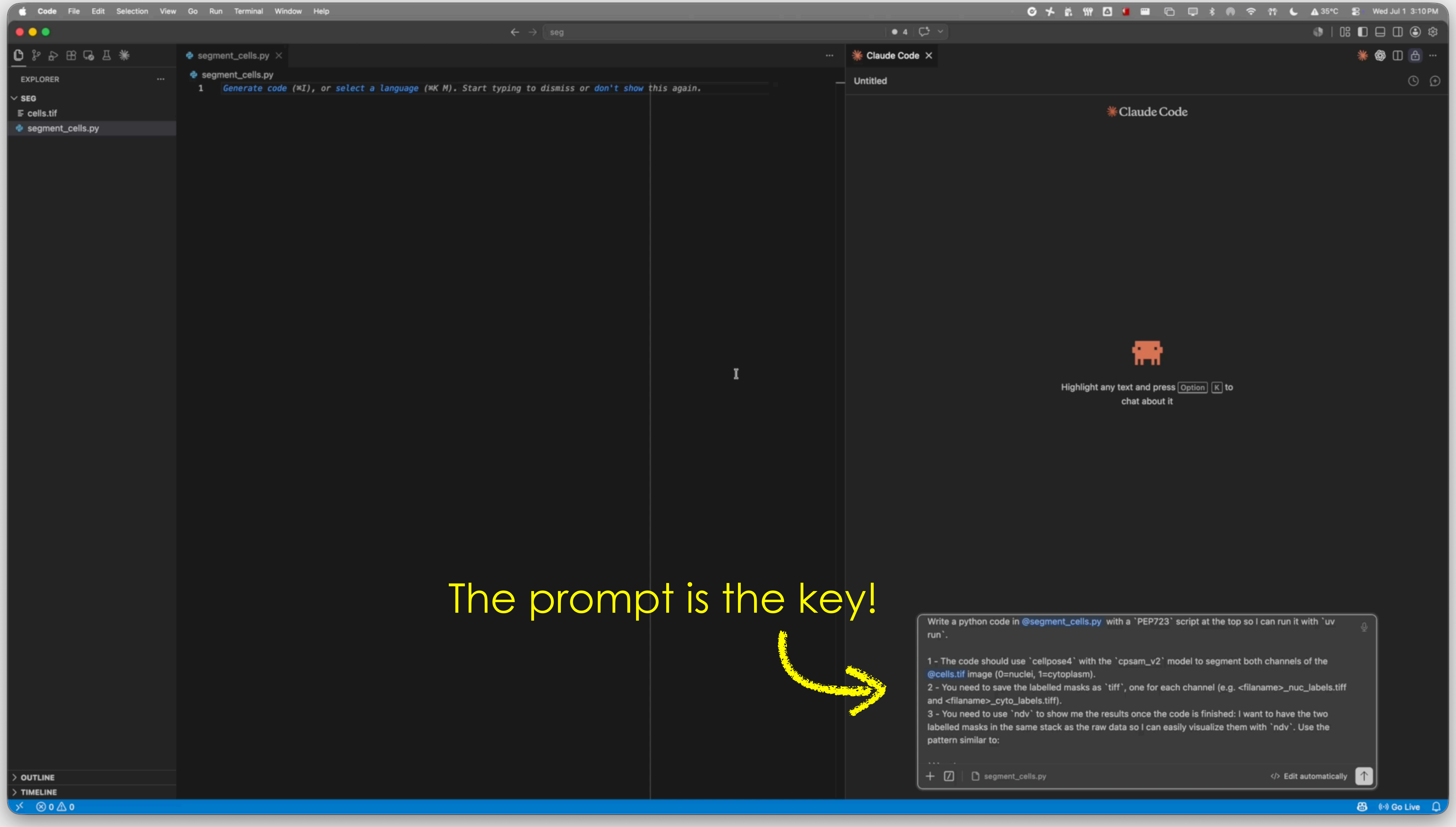

🗣️ Brief Self-Introduction

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1. My **name** is *Federico*
2. My **position** is as a *Director*
3. My **lab** is *the CITE@HMS in Boston, MA*
4. My model **system** is *a microscope*
5. I **acquired** my **data** with *any microscope*